

Assessment Tool - 1

Co-1

APPLICATION Based Questions

Name:- M. Pranay

Reg No:- 192525382

Course Code:- ITA0612

Course Name:- Machine Learning.

1) Probability that the selected box was Box C.

sol:- Given that,

Box A : 6 Red, 4 Blue, 2 Green $\rightarrow$ total = 12

Box B : 5 Red, 7 Blue, 3 Green $\rightarrow$ total = 15

Box C : 3 Red, 2 Blue, 5 Green $\rightarrow$ total = 10

Since B is twice as likely as A and C:

$$P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$$

Probability of blue

$$P(\text{Blue} | A) = \frac{4}{12} = \frac{1}{3}$$

$$P(\text{Blue} | B) = \frac{7}{15}$$

$$P(\text{Blue} | C) = \frac{2}{10} = \frac{1}{5}$$

Using Bayes Theorem:-

$$P(C | \text{Blue}) = \frac{P(C) \cdot P(\text{Blue} | C)}{P(A) P(\text{Blue} | A) + P(B) P(\text{Blue} | B) + P(C) P(\text{Blue} | C)}$$

$$= \frac{\frac{1}{4} \cdot \frac{1}{5}}{\frac{1}{4} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{7}{15} + \frac{1}{4} \cdot \frac{1}{5}}$$

$$= \frac{2}{21}$$

$$P(C | \text{Blue}) = 0.0952$$

2) Probability that the car came from section C.

sol:- Section A: 5 Red, 3 Blue, 2 Black $\rightarrow$ total = 10

Section B: 4 Red, 6 Blue, 5 Black $\rightarrow$ total = 15

Section C: 7 Red, 2 Blue, 6 Black $\rightarrow$ total = 15

B is twice as likely:-

$$P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$$

Blue Probabilities:-

$$P(\text{Blue}/A) = \frac{3}{10}$$

$$P(\text{Blue}/B) = \frac{6}{15} = \frac{2}{5}$$

$$P(\text{Blue}/C) = \frac{2}{15}$$

Therefore,

$$P(C/\text{Blue}) = \frac{\frac{1}{4} \cdot \frac{2}{15}}{\frac{1}{4} \cdot \frac{3}{10} + \frac{1}{2} \cdot \frac{2}{5} + \frac{1}{4} \cdot \frac{2}{15}}$$

$$= \frac{4}{37}$$

$$P(C/\text{Blue}) = 0.1081$$

$$= 10.81\%$$

3) Medical Image Classification:-

The test set contains:-

$$1800 + 1800 = 3600$$

20% is used for testing;

$$3600 \times 0.2 = 720$$

Each class has

$$1800 \times 0.2 = 360$$

Given:

$$TP = 210$$

$$TN = 190$$

$$FP = 30$$

Therefore:

$$FN = 360 - 210 = 150$$

Confusion matrix:-

	Actual Positive	Actual Negative
Predicted Positive	TP = 210	FP = 30
Predicted Negative	FN = 150	TN = 190

Accuracy:-

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{210 + 190}{720}$$

$$= 55.56\%$$

Correction:- Using the stated 720 test samples,
the accuracy is 55.56%.

Precision:-

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{210}{210+30} = 87.5\%$$

Sensitivity (Recall):-

$$\text{Sensitivity} = \frac{TP}{TP+FN} = \frac{210}{210+150} = 58.33\%$$

Specificity:-

$$\text{Specificity} = \frac{TN}{TN+FP} = \frac{190}{190+30} = 86.36\%$$

7) Industrial Tire classification:-

Given:-

$$TP = 119$$

$$FN = 6$$

$$TN = 120$$

$$FP = 5$$

$$\text{Total} = 250.$$

Accuracy:-

$$\text{Accuracy} = \frac{TP+TN}{\text{Total}} = \frac{119+120}{250} = 95.6\%$$

Precision:-

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{119}{119+5} = 95.97\%$$

Sensitivity:-

$$\text{sensitivity} = \frac{TP}{TP+FN} = \frac{119}{119+6} = 95.2\%$$

Specificity:-

$$\text{Specificity} = \frac{TN}{TN+FP} = \frac{120}{120+5} = 96\%$$

F1-Score:-

$$F1 = \frac{2TP}{2TP+FP+FN} = \frac{238}{238+5+6} = 95.58\%$$

8. Medical Diagnosis:-

Given:-

$$TP = 138$$

$$FN = 12$$

$$TN = 135$$

$$FP = 15$$

$$\text{Total} = 300$$

Accuracy:-

$$\text{Accuracy} = \frac{TP+TN}{\text{Total}} = \frac{138+135}{300} = 91\%$$

Precision:-

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{138}{138+15} = 90.2\%$$

Sensitivity:-

$$\text{sensitivity} = \frac{TP}{TP+FN} = \frac{138}{138+12} = 92\%$$

Specificity:-

$$\text{specificity} = \frac{TN}{TN+FP} = \frac{135}{135+15} = 90\%$$

F1-Score:-

$$F1 = \frac{2TP}{2TP+FP+FN} = \frac{2(138)}{2(138)+15+12} = 91.09\%$$